

Skills Summary

Pythagorean Theorem: Which Side Goes Where?

Use this reference while completing your worksheet or when studying.

Skill 1 — Find the hypotenuse (both legs known)

Rule: $a^2 + b^2 = c^2$, and c is **always** the hypotenuse: the side opposite the right angle, and the longest side of the triangle.

Name the hypotenuse before you substitute. Both givens are legs $\Rightarrow$ *add* the squares and take the square root. The answer must come out *longer* than either given side.

Example: A right triangle has legs of 5 and 12. Find the hypotenuse.

Step 1. Both given sides touch the right angle, so both are legs and the unknown is c — that is the adding direction of the theorem.

Step 2. $c = \sqrt{5^2 + 12^2} = 13$, and 13 is longer than both givens, as a hypotenuse must be.
 $\Rightarrow c = 13$

Try it: A right triangle has legs of 8 and 15. Find the hypotenuse.

check: 17

Skill 2 — Find a missing leg (hypotenuse known)

Rule: when the hypotenuse c is given, the missing leg comes from *subtracting*: $b^2 = c^2 - a^2$.

Check: a leg must be *shorter* than the hypotenuse. If your “leg” comes out longer, you added when you should have subtracted — the hypotenuse landed in a leg slot.

Example: A right triangle has hypotenuse 25 and one leg 7. Find the other leg.

Step 1. The longest side is named as the hypotenuse, so it stands alone on the c side and the unknown is a leg — subtract.

Step 2. $b = \sqrt{25^2 - 7^2} = 24$. Adding instead would give about 25.96, longer than the hypotenuse and therefore impossible. $\Rightarrow b = 24$

Try it: A right triangle has hypotenuse 10 and one leg 6. Find the other leg.

check: 8

Skill 3 — Distance between two points

Rule: the horizontal run and the vertical rise are the two **legs**; the segment joining the points is **always the hypotenuse**, because the run and the rise meet at a right angle.

So a distance question is always the adding direction — there is no version of it that subtracts.

Example: Find the distance between $A(-6, 0)$ and $B(0, 8)$.

Step 1. Sliding across and then up builds a right triangle whose legs are the run and the rise, so $\overline{AB}$ is its hypotenuse and the squares get added.

Step 2. run = 6 and rise = 8, so $AB = \sqrt{6^2 + 8^2} = 10$. $\Rightarrow AB = 10$

Try it: Find the distance between $P(-3, 1)$ and $Q(2, 13)$.

check: 13

(!) **Watch out:** “Longest number given” does not mean “hypotenuse”. Only position decides: the hypotenuse is the side *across from the right angle*. If a problem calls two sides legs, the hypotenuse is the thing you are solving for — add. If it names the hypotenuse, subtract. Then check the size of your answer before you write it down.